

Exercise 7: Financial Forecasting

1. Understand Recursive Algorithms

Recursion : Recursion is a programming technique where a function calls itself to solve smaller instances of the same problem until a base condition is reached.

Advantages of Recursion

- Simplifies complex problems.
- Produces cleaner and more readable code.
- Naturally fits divide-and-conquer and forecasting problems.

2. Setup & Implementation

```
class FinancialForecast {  
    public static double futureValue(double presentValue, double growthRate, int  
years) {  
        if (years == 0) return presentValue;  
  
        return futureValue(presentValue, growthRate, years - 1) * (1 + growthRate);  
    }  
}
```

```
public static void main(String[] args) {  
    double presentValue = 10000;  
    double growthRate = 0.10;  
    int years = 5;
```

```
    double res = futureValue(presentValue, growthRate, years);  
    System.out.printf("Future value after %d years = %.2f", years, res);  
}  
}
```

3. Analysis

<i>Operation</i>	<i>Complexity</i>
Recursive Forecasting	$O(n)$

Optimization Techniques

- Use Memoization to store previously computed values.
- Convert recursion into an iterative approach.
- Apply Dynamic Programming for large datasets.
- Reduce recursive stack usage by using loops when possible.